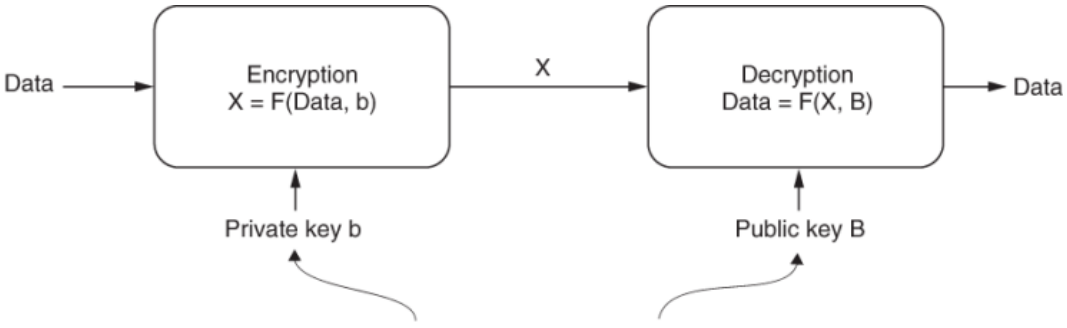
participants. To address this situation, current networked systems use a method in which the keys used for encryption and decryption are different--*asymmetric*. Let's explore the asymmetric key solution and its relevance to the blockchain-based systems.

5.1.2 Asymmetric key cryptography

Asymmetric key cryptography is commonly known as public-key cryptography. This method uses two different keys instead of a single secret key (as in symmetric key cryptography):

- Let `{b, B}` be `{private key, public key}` for a participant in Buffalo, New York, USA.
- Let `{k, K}` be the key pair for a participant in Kathmandu, Nepal.
- Each participant publishes their public key but keeps their private key safe and secure, typically using a passphrase.
- Either participant can use the other's public key to encrypt a message that only that other person can decrypt, using the corresponding secret private key.

The key pair works as shown in figure 5.3. Input data (`Data`) is encrypted using function F and secret private key `b`, resulting in encrypted message X. The message X is decrypted using the same function F, but now with a different key--public key B--to extract the original data.



Encryption and decryption keys are different.